KERALA DIPLOMA CURRICULUM REVISION 2026

Course 4371 | 4 Credits: 4 | Program core | Source-grounded

VLSI Design

- To understand the concept of MOS physics, CMOS logic circuit and design

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 4371 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 4371.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Integrated Circuit Design and Fabrication
Course code	4371
Course title	VLSI Design
Semester	4 Credits: 4
Course category	Program core
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- ● To understand the concept of MOS physics, CMOS logic circuit and design
- ● To understand the MOSFET structure, scaling and fabrication
- ● To understand the VLSI design methodologies

Course outcomes

CO	Official outcome	Study level
CO1	Explain the basic MOS physics 13 Understanding Develop the layout of CMOS logic circuits and	See official syllabus
CO2	15 Understanding explain VLSI design methodologies Demonstrate the realization of CMOS Logic and	See official syllabus
CO3	15 Understanding storage cells Demonstrate the realization of various arithmetic	See official syllabus
CO4	15 Understanding circuits and explain various reliability and testing methods in VLSI	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Explain the basic MOS physics	4	As specified
Module 2	methodologies	4	As specified
Module 3	Demonstrate the realization of CMOS Logic and storage cells	4	As specified
Module 4	reliability and testing methods in VLSI	4	As specified

MODULE 1

Explain the basic MOS physics

This module is presented from the official syllabus scope for VLSI Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the basic MOS physics
- Official duration: As specified
- Official topic entries captured: 4

- The full source excerpt is preserved below for coverage checking.

Explain the basic concepts of MOS capacitor

Explain the basic concepts of MOS capacitor is an official topic in Module 1 of VLSI Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the basic concepts of MOS capacitor using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the basic concepts of mos capacitor should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the basic concepts of MOS capacitor means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Explain the basic concepts of MOS capacitor [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain the structure of MOSFET 2 Understanding Describe various short channel effects in

Explain the structure of MOSFET 2 Understanding Describe various short channel effects in is an official topic in Module 1 of VLSI Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the structure of MOSFET 2 Understanding Describe various short channel effects in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the structure of mosfet 2 understanding describe various short channel effects in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the structure of MOSFET 2 Understanding Describe various short channel effects in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Explain the structure of MOSFET 2 Understanding Describe various short channel effects in □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

4 Understanding MOSFET

4 Understanding MOSFET is an official topic in Module 1 of VLSI Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding MOSFET using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding mosfet should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding MOSFET means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 4 Understanding MOSFET [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain MOSFET capacitance 4 Understanding depletion and inversion, threshold voltage, body effect, MOSFET -Structure, types, Drain current equation (equation only)-linear and saturation region, Drain characteristics, transfer characteristics. Short channel effects (Basic concepts only)-channel length modulation, Drain induced Barrier Lowering, velocity saturation, threshold voltage variations and Hot carrier Effects MOSFET Capacitances-Oxide related capacitance, junction capacitances Develop the layout of CMOS logic circuits and explain VLSI design

Explain MOSFET capacitance 4 Understanding depletion and inversion, threshold voltage, body effect, MOSFET -Structure, types, Drain current equation (equation only)-linear and saturation region, Drain characteristics, transfer characteristics. Short channel effects (Basic concepts only)-channel length modulation, Drain induced Barrier Lowering, velocity saturation, threshold voltage variations and Hot carrier Effects MOSFET Capacitances-Oxide related capacitance, junction capacitances Develop the layout of CMOS logic circuits and explain VLSI design is an official topic in Module 1 of VLSI Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain MOSFET capacitance 4 Understanding depletion and inversion, threshold voltage, body effect, MOSFET -Structure, types, Drain current equation (equation only)-linear and saturation region, Drain characteristics, transfer characteristics. Short channel effects (Basic concepts only)-channel length modulation, Drain induced Barrier Lowering, velocity saturation, threshold voltage variations and Hot carrier Effects MOSFET Capacitances-Oxide related capacitance, junction capacitances Develop the layout of CMOS logic circuits and explain VLSI design using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain mosfet capacitance 4 understanding depletion and inversion, threshold voltage, body effect, mosfet -structure, types, drain current equation (equation only)-linear and saturation region, drain characteristics, transfer characteristics. short channel effects (basic concepts only)-channel length modulation, drain induced barrier lowering, velocity saturation, threshold voltage variations and hot carrier effects mosfet capacitances-oxide related capacitance, junction capacitances develop the layout of cmos logic circuits and explain vlsi design should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain MOSFET capacitance 4 Understanding depletion and inversion, threshold voltage, body effect, MOSFET -Structure, types, Drain current equation (equation only)-linear and saturation region, Drain characteristics, transfer characteristics. Short channel effects (Basic concepts only)-channel length modulation, Drain induced Barrier Lowering, velocity saturation, threshold voltage variations and Hot carrier Effects MOSFET Capacitances-Oxide related capacitance, junction capacitances Develop the layout of CMOS logic circuits and explain VLSI design means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Explain MOSFET capacitance 4 Understanding depletion and inversion, threshold voltage, body effect, MOSFET -Structure, types, Drain current equation (equation only)-linear and saturation region, Drain characteristics, transfer characteristics. Short channel effects (Basic concepts only)-channel length modulation, Drain induced Barrier Lowering, velocity saturation, threshold voltage variations and Hot carrier Effects MOSFET Capacitances-Oxide related capacitance, junction capacitances Develop the layout of CMOS logic circuits and explain VLSI design □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Explain the basic MOS physics

   M1.01     Explain the basic concepts of MOS capacitor              3
Understanding

   M1.02     Explain the structure of MOSFET                          2
Understanding

             Describe various short channel effects in
   M1.03                                                              4
Understanding
             MOSFET

   M1.04     Explain MOSFET capacitance                               4
Understanding

 Contents: Ideal MOS Capacitor, Energy band diagram at equilibrium, accumulation,
 depletion and inversion, threshold voltage, body effect, MOSFET -Structure,
types, Drain
 current equation (equation only)-linear and saturation region, Drain
characteristics, transfer
 characteristics.
 Short channel effects (Basic concepts only)-channel length modulation, Drain
induced
 Barrier Lowering, velocity saturation, threshold voltage variations and Hot
carrier Effects
 MOSFET Capacitances-Oxide related capacitance, junction capacitances
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Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

methodologies

This module is presented from the official syllabus scope for VLSI Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: methodologies
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Explain CMOS inverter

Explain CMOS inverter is an official topic in Module 2 of VLSI Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain CMOS inverter using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain cmos inverter should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain CMOS inverter means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Explain CMOS inverter [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Summarize VLSI design flow

Summarize VLSI design flow is an official topic in Module 2 of VLSI Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize VLSI design flow using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize vlsi design flow should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize VLSI design flow means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Summarize VLSI design flow □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Illustrate MOSFET Fabrication 4 Understanding Develop the Layout of CMOS Inverter, NAND

Illustrate MOSFET Fabrication 4 Understanding Develop the Layout of CMOS Inverter, NAND is an official topic in Module 2 of VLSI Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate MOSFET Fabrication 4 Understanding Develop the Layout of CMOS Inverter, NAND using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate mosfet fabrication 4 understanding develop the layout of cmos inverter, nand should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate MOSFET Fabrication 4 Understanding Develop the Layout of CMOS Inverter, NAND means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Illustrate MOSFET Fabrication 4 Understanding Develop the Layout of CMOS Inverter, NAND □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

4 Understanding and NOR circuits static and dynamic power dissipation VLSI Design Methodologies-Introduction to Moore’s law, ASIC-types of ASIC, SoCs, FPGA devices, VLSI Design flow-Design specifications, Behavioral level RTL Logic Design and Physical level design(basics concept only) MOSFET Fabrication Techniques and layout-Twin Tub Fabrication sequence, fabrication process flow, Layout Design an Design rules, Stick Diagram and Design rules- micron rules and Lambda rules(Definition only),layout of CMOS inverter, two input NAND and NOR gates

4 Understanding and NOR circuits static and dynamic power dissipation VLSI Design Methodologies-Introduction to Moore’s law, ASIC-types of ASIC, SoCs, FPGA devices, VLSI Design flow-Design specifications, Behavioral level RTL Logic Design and Physical level design(basics concept only) MOSFET Fabrication Techniques and layout-Twin Tub Fabrication sequence, fabrication process flow, Layout Design an Design rules, Stick Diagram and Design rules- micron rules and Lambda rules(Definition only),layout of CMOS inverter, two input NAND and NOR gates is an official topic in Module 2 of VLSI Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding and NOR circuits static and dynamic power dissipation VLSI Design Methodologies-Introduction to Moore’s law, ASIC-types of ASIC, SoCs, FPGA devices, VLSI Design flow-Design specifications, Behavioral level RTL Logic Design and Physical level design(basics concept only) MOSFET Fabrication Techniques and layout-Twin Tub Fabrication sequence, fabrication process flow, Layout Design an Design rules, Stick Diagram and Design rules- micron rules and Lambda rules(Definition only),layout of CMOS inverter, two input NAND and NOR gates using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding and nor circuits static and dynamic power dissipation vlsi design methodologies-introduction to moore’s law, asic-types of asic, socs, fpga devices, vlsi design flow-design specifications, behavioral level rtl logic design and physical level design(basics concept only) mosfet fabrication techniques and layout-twin tub fabrication sequence, fabrication process flow, layout design an design rules, stick diagram and design rules- micron rules and lambda rules(definition only),layout of cmos inverter, two input nand and nor gates should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding and NOR circuits static and dynamic power dissipation VLSI Design Methodologies-Introduction to Moore’s law, ASIC-types of ASIC, SoCs, FPGA devices, VLSI Design flow-Design specifications, Behavioral level RTL Logic Design and Physical level design(basics concept only) MOSFET Fabrication Techniques and layout-Twin Tub Fabrication sequence, fabrication process flow, Layout Design an Design rules, Stick Diagram and Design rules- micron rules and Lambda rules(Definition only),layout of CMOS inverter, two input NAND and NOR gates means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Understanding and NOR circuits static and dynamic power dissipation VLSI Design Methodologies-Introduction to Moore’s law, ASIC-types of ASIC, SoCs, FPGA devices, VLSI Design flow-Design specifications, Behavioral level RTL Logic Design and Physical level design(basics concept only) MOSFET Fabrication Techniques and layout-Twin Tub Fabrication sequence, fabrication process flow, Layout Design an Design rules, Stick Diagram and Design rules- micron rules and Lambda rules(Definition only),layout of CMOS inverter, two input NAND and NOR gates □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
methodologies
  M2.01    Explain CMOS inverter                                   3
Understanding

  M2.02    Summarize VLSI design flow                              4
Understanding

  M2.03    Illustrate MOSFET Fabrication                           4
Understanding
           Develop the Layout of CMOS Inverter, NAND
  M2.04                                                            4
Understanding
           and NOR circuits

           Series Test 1                                           1

Contents: CMOS inverter-Switching characteristics, Noise margin, propogation
delay,
static and dynamic power dissipation
VLSI Design Methodologies-Introduction to Moore’s law, ASIC-types of ASIC, SoCs,
FPGA devices, VLSI Design flow-Design specifications, Behavioral level RTL Logic
Design and Physical level design(basics concept only)
MOSFET Fabrication Techniques and layout-Twin Tub Fabrication sequence,
fabrication process flow, Layout Design an Design rules, Stick Diagram and Design
rules-
micron rules and Lambda rules(Definition only),layout of CMOS inverter, two input
NAND
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Demonstrate the realization of CMOS Logic and storage cells

This module is presented from the official syllabus scope for VLSI Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Demonstrate the realization of CMOS Logic and storage cells
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

3 Understanding techniques Explain various Dynamic CMOS Logic Design

3 Understanding techniques Explain various Dynamic CMOS Logic Design is an official topic in Module 3 of VLSI Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding techniques Explain various Dynamic CMOS Logic Design using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding techniques explain various dynamic cmos logic design should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding techniques Explain various Dynamic CMOS Logic Design means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 3 Understanding techniques Explain various Dynamic CMOS Logic Design □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

4 Understanding techniques Explain the realization of ROM arrays based on

4 Understanding techniques Explain the realization of ROM arrays based on is an official topic in Module 3 of VLSI Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding techniques Explain the realization of ROM arrays based on using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding techniques explain the realization of rom arrays based on should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding techniques Explain the realization of ROM arrays based on means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 4 Understanding techniques Explain the realization of ROM arrays based on □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

4 Understanding NOR and NAND logic

4 Understanding NOR and NAND logic is an official topic in Module 3 of VLSI Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding NOR and NAND logic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding nor and nand logic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding NOR and NAND logic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 4 Understanding NOR and NAND logic □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Explain the realization of SRAM and DRAM 4 Understanding logic, Pass transistor logic and transmission gate logic. Dynamic Logic Design and storage cells-Pre-charge-Evaluate logic, Domino Logic,NP domino logic, Read Only Memory-4x4 MOS ROM cell Arrays(OR,NOR,NAND),Random Access Memory-SRAM-Six transistor CMOS SRAM cell, DRAM-Three transistor and one transistor Dynamic memory cell Demonstrate the realization of various arithmetic circuits and explain various

Explain the realization of SRAM and DRAM 4 Understanding logic, Pass transistor logic and transmission gate logic. Dynamic Logic Design and storage cells-Pre-charge-Evaluate logic, Domino Logic,NP domino logic, Read Only Memory-4x4 MOS ROM cell Arrays(OR,NOR,NAND),Random Access Memory-SRAM-Six transistor CMOS SRAM cell, DRAM-Three transistor and one transistor Dynamic memory cell Demonstrate the realization of various arithmetic circuits and explain various is an official topic in Module 3 of VLSI Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the realization of SRAM and DRAM 4 Understanding logic, Pass transistor logic and transmission gate logic. Dynamic Logic Design and storage cells-Pre-charge-Evaluate logic, Domino Logic,NP domino logic, Read Only Memory-4x4 MOS ROM cell Arrays(OR,NOR,NAND),Random Access Memory-SRAM-Six transistor CMOS SRAM cell, DRAM-Three transistor and one transistor Dynamic memory cell Demonstrate the realization of various arithmetic circuits and explain various using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the realization of sram and dram 4 understanding logic, pass transistor logic and transmission gate logic. dynamic logic design and storage cells-pre-charge-evaluate logic, domino logic,np domino logic, read only memory-4x4 mos rom cell arrays(or,nor,nand),random access memory-sram-six transistor cmos sram cell, dram-three transistor and one transistor dynamic memory cell demonstrate the realization of various arithmetic circuits and explain various should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the realization of SRAM and DRAM 4 Understanding logic, Pass transistor logic and transmission gate logic. Dynamic Logic Design and storage cells-Pre-charge-Evaluate logic, Domino Logic,NP domino logic, Read Only Memory-4x4 MOS ROM cell Arrays(OR,NOR,NAND),Random Access Memory-SRAM-Six transistor CMOS SRAM cell, DRAM-Three transistor and one transistor Dynamic memory cell Demonstrate the realization of various arithmetic circuits and explain various means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Explain the realization of SRAM and DRAM 4 Understanding logic, Pass transistor logic and transmission gate logic. Dynamic Logic Design and storage cells-Pre-charge-Evaluate logic, Domino Logic,NP domino logic, Read Only Memory-4x4 MOS ROM cell Arrays(OR,NOR,NAND),Random Access Memory-SRAM-Six transistor CMOS SRAM cell, DRAM-Three transistor and one transistor Dynamic memory cell Demonstrate the realization of various arithmetic circuits and explain various □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Demonstrate the realization of CMOS Logic and storage cells

           Explain various Static CMOS Logic Design
  M3.01                                                           3
Understanding
           techniques
           Explain various Dynamic CMOS Logic Design
  M3.02                                                           4
Understanding
           techniques
           Explain the realization of ROM arrays based on
  M3.03                                                           4
Understanding
           NOR and NAND logic

  M3.04    Explain the realization of SRAM and DRAM               4
Understanding

Contents: Static CMOS Logic Design-Realization of logic functions with static
CMOS
logic, Pass transistor logic and transmission gate logic.
Dynamic Logic Design and storage cells-Pre-charge-Evaluate logic, Domino Logic,NP
domino logic, Read Only Memory-4x4 MOS ROM cell Arrays(OR,NOR,NAND),Random
Access Memory-SRAM-Six transistor CMOS SRAM cell, DRAM-Three transistor and one
transistor Dynamic memory cell

           Demonstrate the realization of various arithmetic circuits and explain
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

reliability and testing methods in VLSI

This module is presented from the official syllabus scope for VLSI Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: reliability and testing methods in VLSI
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Explain the realization of various adder circuits

Explain the realization of various adder circuits is an official topic in Module 4 of VLSI Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the realization of various adder circuits using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the realization of various adder circuits should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the realization of various adder circuits means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Explain the realization of various adder circuits [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain the realization of Multipliers

Explain the realization of Multipliers is an official topic in Module 4 of VLSI Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the realization of Multipliers using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the realization of multipliers should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the realization of Multipliers means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Explain the realization of Multipliers [illegible] definition/meaning [illegible]. [illegible], steps, application [illegible]. Technical terms English-[illegible] [illegible].

Explain Sense amplifiers 3 Understanding Outline various reliability and testing methods in

Explain Sense amplifiers 3 Understanding Outline various reliability and testing methods in is an official topic in Module 4 of VLSI Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Sense amplifiers 3 Understanding Outline various reliability and testing methods in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain sense amplifiers 3 understanding outline various reliability and testing methods in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Sense amplifiers 3 Understanding Outline various reliability and testing methods in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Explain Sense amplifiers 3 Understanding Outline various reliability and testing methods in □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

4 Understanding VLSI select adder, square root carry-select adder, multipliers-Array multiplier Sense Amplifiers-Basic principles of sense amplifiers, differential sense amplifiers Reliability and testing of VLSI Circuits-General concept, MOS testing, test generation methods-Built-In-Self-Test (BIST) and Automatic Test Pattern Generation (ATPG) Text / Reference: T/R Book Title/Author T1 Sung –Mo Kang & Yusuf Leblebici, CMOS Digital Integrated Circuits- Analysis & Design, McGraw-Hill, Third Ed., 2003 T2 Wayne Wolf ,Modern VLSI design, Third Edition, Pearson Education,2002 R1 Neil H.E. Weste, Kamran Eshraghian, Principles of CMOS VLSI Design- A Systems Perspective, Second Edition. Pearson Publication, 2005 R2 Razavi - Design of Analog CMOS Integrated Circuits,1e, McGraw Hill Education India Education, New Delhi, 2003 R3 S.M. SZE, VLSI Technology, 2/e, Indian Edition, McGraw-Hill,2003 R4 Suresh Babu V, Solid State Device Technology, Sanguine, 2005. Online Resources: Sl.No. Website Link 1 http://ece-research.unm.edu/jimp/vlsi/slides/chap5_2.html 2 https://www.elprocus.com/cmos-inverter/#:~:text=CMOS 3 https://www.rohm.com/electronics-basics/transistors/understanding-mosfet-characteristics

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Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding VLSI select adder, square root carry-select adder, multipliers-Array multiplier Sense Amplifiers-Basic principles of sense amplifiers, differential sense amplifiers Reliability and testing of VLSI Circuits-General concept, MOS testing, test generation methods-Built-In-Self-Test (BIST) and Automatic Test Pattern Generation (ATPG) Text / Reference: T/R Book Title/Author T1 Sung –Mo Kang & Yusuf Leblebici, CMOS Digital Integrated Circuits- Analysis & Design, McGraw-Hill, Third Ed., 2003 T2 Wayne Wolf ,Modern VLSI design, Third Edition, Pearson Education,2002 R1 Neil H.E. Weste, Kamran Eshraghian, Principles of CMOS VLSI Design- A Systems Perspective, Second Edition. Pearson Publication, 2005 R2 Razavi - Design of Analog CMOS Integrated Circuits,1e, McGraw Hill Education India Education, New Delhi, 2003 R3 S.M. SZE, VLSI Technology, 2/e, Indian Edition, McGraw-Hill,2003 R4 Suresh Babu V, Solid State Device Technology, Sanguine, 2005. Online Resources: Sl.No. Website Link 1 http://ece-research.unm.edu/jimp/vlsi/slides/chap5_2.html 2 https://www.elprocus.com/cmos-inverter/#:~:text=CMOS 3 https://www.rohm.com/electronics-basics/transistors/understanding-mosfet- characteristics using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding vlsi select adder, square root carry-select adder, multipliers-array multiplier sense amplifiers-basic principles of sense amplifiers, differential sense amplifiers reliability and testing of vlsi circuits-general concept, mos testing, test generation methods-built-in-self-test (bist) and automatic test pattern generation (atpg) text / reference: t/r book title/author t1 sung –mo kang & yusuf leblebici, cmos digital integrated circuits- analysis & design, mcgraw-hill, third ed., 2003 t2 wayne wolf ,modern vlsi design, third edition, pearson education,2002 r1 neil h.e. weste, kamran eshraghian, principles of cmos vlsi design- a systems perspective, second edition. pearson publication, 2005 r2 razavi - design of analog cmos integrated circuits,1e, mcgraw hill education india education, new delhi, 2003 r3 s.m. sze, vlsi technology, 2/e, indian edition, mcgraw-hill,2003 r4 suresh babu v, solid state device technology, sanguine, 2005. online resources: sl.no. website link 1 http://ece-research.unm.edu/jimp/vlsi/slides/chap5_2.html 2 https://www.elprocus.com/cmos-inverter/#:~:text=cmos 3 https://www.rohm.com/electronics-basics/transistors/understanding-mosfet- characteristics should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding VLSI select adder, square root carry-select adder, multipliers-Array multiplier Sense Amplifiers-Basic principles of sense amplifiers, differential sense amplifiers Reliability and testing of VLSI Circuits-General concept, MOS testing, test generation methods-Built-In-Self-Test (BIST) and Automatic Test Pattern Generation (ATPG) Text / Reference: T/R Book Title/Author T1 Sung –Mo Kang & Yusuf Leblebici, CMOS Digital Integrated Circuits- Analysis & Design, McGraw-Hill, Third Ed., 2003 T2 Wayne Wolf ,Modern VLSI design, Third Edition, Pearson Education,2002 R1 Neil H.E. Weste, Kamran Eshraghian, Principles of CMOS VLSI Design- A Systems Perspective, Second Edition. Pearson Publication, 2005 R2 Razavi - Design of Analog CMOS Integrated Circuits,1e, McGraw Hill Education India Education, New Delhi, 2003 R3 S.M. SZE, VLSI Technology, 2/e, Indian Edition, McGraw-Hill,2003 R4 Suresh Babu V, Solid State Device Technology, Sanguine, 2005. Online Resources: Sl.No. Website Link 1 http://ece-research.unm.edu/jimp/vlsi/slides/chap5_2.html 2 https://www.elprocus.com/cmos-inverter/#:~:text=CMOS 3 https://www.rohm.com/electronics-basics/transistors/understanding-mosfet- characteristics means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 4 Understanding VLSI select adder, square root carry-select adder, multipliers-Array multiplier Sense Amplifiers-Basic principles of sense amplifiers, differential sense amplifiers Reliability and testing of VLSI Circuits-General concept, MOS testing, test generation methods-Built-In-Self-Test (BIST) and Automatic Test Pattern Generation (ATPG) Text / Reference: T/R Book Title/Author T1 Sung –Mo Kang & Yusuf Leblebici, CMOS Digital Integrated Circuits- Analysis & Design, McGraw-Hill, Third Ed., 2003 T2 Wayne Wolf ,Modern VLSI design, Third Edition, Pearson Education,2002 R1 Neil H.E. Weste, Kamran Eshraghian, Principles of CMOS VLSI Design- A Systems Perspective, Second Edition. Pearson Publication, 2005 R2 Razavi - Design of Analog CMOS Integrated Circuits,1e, McGraw Hill Education India Education, New Delhi, 2003 R3 S.M. SZE, VLSI Technology, 2/e, Indian Edition, McGraw-Hill,2003 R4 Suresh Babu V, Solid State Device Technology, Sanguine, 2005. Online Resources: Sl.No. Website Link 1 http://ece-research.unm.edu/jimp/vlsi/slides/chap5_2.html 2 https://www.elprocus.com/cmos-inverter/#:~:text=CMOS 3 https://www.rohm.com/electronics-basics/transistors/understanding-mosfet- characteristics □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
reliability and testing methods in VLSI

  M4.01    Explain the realization of various adder circuits      4
Understanding

  M4.02    Explain the realization of Multipliers                 4
Understanding
   M4.03      Explain Sense amplifiers                                 3
Understanding

              Outline various reliability and testing methods in
   M4.04                                                               4
Understanding
              VLSI
              Series Test 2                                            1

 Contents: Arithmetic Circuits-Adders-Static Adder, Carry-By pass adder, Linear
carry-
 select adder, square root carry-select adder, multipliers-Array multiplier
 Sense Amplifiers-Basic principles of sense amplifiers, differential sense
amplifiers
 Reliability and testing of VLSI Circuits-General concept, MOS testing, test
generation
 methods-Built-In-Self-Test (BIST) and Automatic Test Pattern Generation (ATPG)

Text / Reference:
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

> **Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Explain the basic concepts of MOS capacitor. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the basic concepts of MOS capacitor*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain the structure of MOSFET 2 Understanding Describe various short channel effects in. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the structure of MOSFET 2 Understanding Describe various short channel effects in*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding MOSFET. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding MOSFET*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain MOSFET capacitance 4 Understanding depletion and inversion, threshold voltage, body effect, MOSFET -Structure, types, Drain current equation (equation only)-linear and saturation region, Drain characteristics, transfer characteristics. Short channel effects (Basic concepts only)-channel length modulation, Drain induced Barrier Lowering, velocity saturation, threshold voltage variations and Hot carrier Effects MOSFET

Capacitances-Oxide related capacitance, junction capacitances Develop the layout of CMOS logic circuits and explain VLSI design. (3 marks)

Answer: Begin with the standard definition or identity of *Explain MOSFET capacitance 4 Understanding depletion and inversion, threshold voltage, body effect, MOSFET - Structure, types, Drain current equation (equation only)-linear and saturation region, Drain characteristics, transfer characteristics. Short channel effects (Basic concepts only)-channel length modulation, Drain induced Barrier Lowering, velocity saturation, threshold voltage variations and Hot carrier Effects MOSFET Capacitances-Oxide related capacitance, junction capacitances Develop the layout of CMOS logic circuits and explain VLSI design*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain Explain CMOS inverter. (3 marks)

Answer: Begin with the standard definition or identity of *Explain CMOS inverter*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Summarize VLSI design flow. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize VLSI design flow*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Illustrate MOSFET Fabrication 4 Understanding Develop the Layout of CMOS Inverter, NAND. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate MOSFET Fabrication 4 Understanding Develop the Layout of CMOS Inverter, NAND*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding and NOR circuits static and dynamic power dissipation VLSI Design Methodologies-Introduction to Moore’s law, ASIC-types of ASIC, SoCs, FPGA devices, VLSI Design flow-Design specifications, Behavioral level RTL Logic Design and Physical level design(basics concept only) MOSFET Fabrication Techniques and layout-Twin Tub Fabrication sequence, fabrication process flow, Layout Design an Design rules, Stick Diagram and Design rules- micron rules and Lambda rules(Definition only),layout of CMOS inverter, two input NAND and NOR gates. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding and NOR circuits static and dynamic power dissipation VLSI Design Methodologies-Introduction to Moore’s law, ASIC-types of ASIC, SoCs, FPGA devices, VLSI Design flow-Design specifications, Behavioral level RTL Logic Design and Physical level design(basics concept only) MOSFET Fabrication Techniques and layout-Twin Tub Fabrication sequence, fabrication process flow, Layout Design an Design rules, Stick Diagram and Design rules- micron rules and Lambda rules(Definition only),layout of CMOS inverter, two input NAND and NOR gates*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain 3 Understanding techniques Explain various Dynamic CMOS Logic Design. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding techniques Explain various Dynamic CMOS Logic Design*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding techniques Explain the realization of ROM arrays based on. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding techniques Explain the realization of ROM arrays based on*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding NOR and NAND logic. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding NOR and NAND logic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain the realization of SRAM and DRAM 4 Understanding logic, Pass transistor logic and transmission gate logic. Dynamic Logic Design and storage cells-Pre-charge-Evaluate logic, Domino Logic,NP domino logic,

Read Only Memory-4x4 MOS ROM cell Arrays(OR,NOR,NAND),Random Access Memory-SRAM-Six transistor CMOS SRAM cell, DRAM-Three transistor and one transistor Dynamic memory cell Demonstrate the realization of various arithmetic circuits and explain various. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the realization of SRAM and DRAM 4 Understanding logic, Pass transistor logic and transmission gate logic. Dynamic Logic Design and storage cells-Pre-charge-Evaluate logic, Domino Logic,NP domino logic, Read Only Memory-4x4 MOS ROM cell Arrays(OR,NOR,NAND),Random Access Memory-SRAM-Six transistor CMOS SRAM cell, DRAM-Three transistor and one transistor Dynamic memory cell Demonstrate the realization of various arithmetic circuits and explain various*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Module 4

Define or explain Explain the realization of various adder circuits. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the realization of various adder circuits*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Explain the realization of Multipliers. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the realization of Multipliers*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain Sense amplifiers 3 Understanding Outline various reliability and testing methods in. (3 marks)

Answer: Begin with the standard definition or identity of *Explain Sense amplifiers 3 Understanding Outline various reliability and testing methods in*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding VLSI select adder, square root carry-select adder, multipliers-Array multiplier Sense Amplifiers-Basic principles of sense amplifiers, differential sense amplifiers Reliability and testing of VLSI Circuits-General concept, MOS testing, test generation methods-Built-In-Self-Test (BIST) and Automatic Test Pattern Generation (ATPG) Text / Reference: T/R Book Title/Author T1 Sung –Mo Kang & Yusuf Leblebici, CMOS Digital Integrated Circuits- Analysis & Design, McGraw-Hill, Third Ed., 2003 T2 Wayne Wolf ,Modern VLSI design, Third Edition, Pearson Education,2002 R1 Neil H.E. Weste, Kamran Eshraghian, Principles of CMOS VLSI Design- A Systems Perspective, Second Edition. Pearson Publication, 2005 R2 Razavi - Design of Analog CMOS Integrated Circuits,1e, McGraw Hill Education India Education, New Delhi, 2003 R3 S.M. SZE, VLSI Technology, 2/e, Indian Edition, McGraw-Hill,2003 R4 Suresh Babu V, Solid State Device Technology, Sanguine, 2005. Online Resources: Sl.No. Website Link 1 http://ece-research.unm.edu/jimp/vlsi/slides/chap5_2.html 2 https://www.elprocus.com/cmos-inverter/#:~:text=CMOS 3 https://www.rohm.com/electronics-basics/transistors/understanding-mosfet-characteristics. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding VLSI select adder, square root carry-select adder, multipliers-Array multiplier Sense Amplifiers-Basic principles of sense amplifiers, differential sense amplifiers Reliability and testing of VLSI Circuits-General concept, MOS testing, test generation methods-Built-In-Self-Test (BIST) and Automatic Test Pattern Generation (ATPG) Text / Reference: T/R Book Title/Author T1 Sung –Mo Kang & Yusuf Leblebici, CMOS Digital Integrated Circuits- Analysis & Design,* *McGraw-Hill, Third Ed., 2003 T2 Wayne Wolf ,Modern VLSI design, Third Edition, Pearson Education,2002 R1 Neil H.E. Weste, Kamran Eshraghian, Principles of CMOS VLSI Design-A Systems Perspective, Second Edition. Pearson Publication, 2005 R2 Razavi - Design of Analog CMOS Integrated Circuits,1e, McGraw Hill Education India Education, New Delhi, 2003 R3 S.M. SZE, VLSI Technology, 2/e, Indian Edition, McGraw-Hill,2003 R4 Suresh Babu V, Solid State Device Technology, Sanguine, 2005. Online Resources: Sl.No. Website Link 1 http://ece-research.unm.edu/jimp/vlsi/slides/chap5_2.html 2 https://www.elprocus.com/cmos-inverter/#:~:text=CMOS 3 https://www.rohm.com/electronics-basics/transistors/understanding-mosfet-characteristics*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Explain the basic concepts of MOS capacitor**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Explain CMOS inverter**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **3 Understanding techniques Explain various Dynamic CMOS Logic Design**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Explain the realization of various adder circuits**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.

- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No. Website Link
- http://ece-research.unm.edu/jimp/vlsi/slides/chap5_2.html
- https://www.elprocus.com/cmos-inverter/#:~:text=CMOS
- https://www.rohm.com/electronics-basics/transistors/understanding-mosfet-characteristics

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 4371. Verify institutional updates against the current official curriculum.